



UNIT-IV

INVERSE LAPLACE TRANSFORM

Topic Learning Objectives:

Upon Completion of this unit, students will be able to:

- Obtain inverse Laplace transform of rational functions.
- Solve ordinary linear differential equations

Definition

Let $L\{f(t)\} = F(s)$, then $f(t)$ is defined as the inverse Laplace transform of $F(s)$ and is denoted by $L^{-1}\{F(s)\}$. Thus $L^{-1}\{F(s)\} = f(t)$.

Linearity Property

Let $L^{-1}\{F(s)\} = f(t)$ and $L^{-1}\{G(s)\} = g(t)$ and a and b be any two constants, Then $L^{-1}[aF(s) + bG(s)] = aL^{-1}\{F(s)\} + bL^{-1}\{G(s)\}$

Table of Inverse Laplace Transforms

$F(s)$	$f(t) = L^{-1}\{F(s)\}$
$\frac{1}{s}, s > 0$	1
$\frac{1}{s-a}, s > a$	e^{at}
$\frac{s}{s^2+a^2}, s > 0$	$\cos at$
$\frac{1}{s^2+a^2}, s > 0$	$\frac{\sin at}{a}$
$\frac{1}{s^2-a^2}, s > a $	$\frac{\sinh at}{a}$
$\frac{s}{s^2-a^2}, s > a $	$\cosh at$
$\frac{1}{s^{n+1}}, s > 0$ $n = 0, 1, 2, 3, \dots$	$\frac{t^n}{n!}$
$\frac{1}{s^{n+1}}, s > 0$ $n \neq -1, -2, -3, \dots$	$\frac{t^n}{\Gamma(n+1)}$

**Examples****1.** Find the inverse Laplace transforms of the following:

$$(i) \frac{1}{2s-5} \quad (ii) \frac{s+b}{s^2+a^2} \quad (iii) \frac{2s-5}{4s^2+25} + \frac{4s-9}{9-s^2}$$

Solution:

$$(i) \quad L^{-1}\left\{\frac{1}{2s-5}\right\} = \frac{1}{2} L^{-1}\left\{\frac{1}{s-\frac{5}{2}}\right\} = \frac{1}{2} e^{\frac{5t}{2}}$$

$$(ii) \quad L^{-1}\left\{\frac{s+b}{s^2+a^2}\right\} = L^{-1}\left\{\frac{s}{s^2+a^2}\right\} + b L^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \cos at + \frac{b}{a} \sin at$$

$$(iii) \quad L^{-1}\left[\frac{2s-5}{4s^2+25} + \frac{4s-8}{9-s^2}\right] = \frac{2}{4} L^{-1}\left\{\frac{s-\frac{5}{2}}{s^2+\frac{25}{4}}\right\} - 4 L^{-1}\left\{\frac{s-\frac{9}{2}}{s^2-9}\right\}$$

$$= \frac{1}{2} \left[\cos \frac{5t}{2} - \sin \frac{5t}{2} \right] - 4 \left[\cos h3t - \frac{3}{2} \sin h3t \right]$$

Shifting property [Evaluation of $L^{-1}[F(s-a)]$]If $L^{-1}[F(s)] = f(t)$, then $L^{-1}[F(s-a)] = e^{at} L^{-1}[F(s)] = e^{at} f(t)$ **Examples:**

$$\mathbf{1. Evaluate: } L^{-1}\left\{\frac{3s+1}{(s+1)^4}\right\}.$$

Solution:

$$\begin{aligned} L^{-1}\left\{\frac{3(s+1-1)+1}{(s+1)^4}\right\} &= 3 L^{-1}\left\{\frac{1}{(s+1)^3}\right\} - 2 L^{-1}\left\{\frac{1}{(s+1)^4}\right\} \\ &= 3e^{-t} L^{-1}\left\{\frac{1}{s^3}\right\} - 2e^{-t} L^{-1}\left\{\frac{1}{s^4}\right\} \end{aligned}$$

Using the formula

$$L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!} \quad \text{and taking } n=2 \text{ and } 3, \text{ we get}$$

$$= \frac{3e^{-t}t^2}{2} - \frac{e^{-t}t^3}{3}.$$

2. Evaluate: $L^{-1}\left\{\frac{s+2}{s^2-2s+5}\right\}$.

Solution:

$$\begin{aligned}\text{Given} &= L^{-1}\left\{\frac{s+2}{(s-1)^2+4}\right\} = L^{-1}\left\{\frac{(s-1)+3}{(s-1)^2+4}\right\} = L^{-1}\left\{\frac{s-1}{(s-1)^2+4}\right\} + 3L^{-1}\left\{\frac{1}{(s-1)^2+4}\right\} \\ &= e^t L^{-1}\left\{\frac{s}{s^2+4}\right\} + 3e^t L^{-1}\left\{\frac{1}{s^2+4}\right\} \\ &= e^t \cos 2t + \frac{3}{2} e^t \sin 2t\end{aligned}$$

3. Evaluate: $L^{-1}\left\{\frac{2s+1}{s^2+3s+1}\right\}$.

Solution:

$$\begin{aligned}\text{Given} &= 2L^{-1}\left\{\frac{\left(s+\frac{3}{2}\right)-1}{\left(s+\frac{3}{2}\right)^2-\frac{5}{4}}\right\} = 2\left[L^{-1}\left\{\frac{\left(s+\frac{3}{2}\right)}{\left(s+\frac{3}{2}\right)^2-\frac{5}{4}}\right\} - L^{-1}\left\{\frac{1}{\left(s+\frac{3}{2}\right)^2-\frac{5}{4}}\right\}\right] \\ &= 2\left[e^{\frac{-3t}{2}} L^{-1}\left\{\frac{s}{s^2-\frac{5}{4}}\right\} - e^{\frac{-3t}{2}} L^{-1}\left\{\frac{1}{s^2-\frac{5}{4}}\right\}\right] \\ &= 2e^{\frac{-3t}{2}} \left[\cos\left[\frac{\sqrt{5}}{2}t\right] - \frac{2}{\sqrt{5}} \sin\left[\frac{\sqrt{5}}{2}t\right]\right].\end{aligned}$$

4. Evaluate: $L^{-1}\left\{\frac{2s^2+5s-4}{s^3+s^2-2s}\right\}$.

Solution:

$$\text{We have } \frac{2s^2+5s-4}{s^3+s^2-2s} = \frac{2s^2+5s-4}{s(s^2+s-2)} = \frac{2s^2+5s-4}{s(s+2)(s-1)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s-1} \quad \text{----- (1)}$$

$$\text{Then } 2s^2+5s-4 = A(s+2)(s-1) + Bs(s-1) + Cs(s+2)$$

For $s = 0$, $A = 2$, for $s = 1$, $C = 1$ and for $s = -2$, $B = -1$.

Using these values in (1),

$$L^{-1}\left\{\frac{2s^2+5s-4}{s^3+s^2-2s}\right\}=L^{-1}\left\{\frac{2}{s}\right\}-L^{-1}\left\{\frac{1}{s+2}\right\}+L^{-1}\left\{\frac{1}{s-1}\right\}=2-e^{-2t}+e^t$$

5. Use the method of partial fractions to find the time signals corresponding to the Laplace transform function

$$F(s) = \frac{4s+5}{(s+1)^2(s+2)}$$

Solution:

Consider

$$\frac{4s+5}{(s+1)^2(s+2)} = \frac{A}{(s+1)^2} + \frac{B}{s+1} + \frac{C}{s+2}$$

$$\text{Then } 4s+5 = A(s+2) + B(s+1)(s+2) + C(s+1)^2$$

$$\text{For } s = -1, A = 1, \text{ for } s = -2, C = -3$$

Comparing the coefficients of s^2 to get $B + C = 0$ so that $B = 3$.

Using these values in (1) to get

$$\frac{4s+5}{(s+1)^2(s+2)} = \frac{1}{(s+1)^2} + \frac{3}{s+1} - \frac{3}{s+2}$$

Hence

$$\begin{aligned} L^{-1}\left\{\frac{4s+5}{(s+1)^2(s+2)}\right\} &= e^{-t}L^{-1}\left\{\frac{1}{s^2}\right\} + 3e^{-t}L^{-1}\left\{\frac{1}{s}\right\} - 3e^{-2t}L^{-1}\left\{\frac{1}{s}\right\} \\ &= te^{-t} + 3e^{-t} - 3e^{-2t} \end{aligned}$$

$$\text{5. Evaluate: } L^{-1}\left\{\frac{s^3}{s^4-a^4}\right\}.$$

Solution:

$$\text{Let } \frac{s^3}{s^4-a^4} = \frac{A}{s-a} + \frac{B}{s+a} + \frac{Cs+D}{s^2+a^2} \quad \text{-----} \quad (1)$$

$$\text{Hence, } s^3 = A(s+a)(s^2+a^2) + B(s-a)(s^2+a^2) + (Cs+D)(s^2-a^2)$$

For $s = a, A = 1/4$, for $s = -a, B = 1/4$, comparing the constant terms to get

$$D = a, (A-B) = 0,$$

Comparing the coefficients of s^3 to get $1 = A + B + C$ and so $C = 1/2$.

Using these values in (1),

$$\frac{s^3}{s^4-a^4} = \frac{1}{4}\left[\frac{1}{s-a} + \frac{1}{s+a}\right] + \frac{1}{2}\frac{s}{s^2+a^2}$$

Taking inverse transforms,

$$L^{-1}\left\{\frac{s^3}{s^4 - a^4}\right\} = \frac{1}{4}[e^{at} + e^{-at}] + \frac{1}{2}\cos at = \frac{1}{2}[\cos at + \cos at]$$

6. Evaluate: $L^{-1}\left\{\frac{s}{s^4 + s^2 + 1}\right\}$.

Solution:

$$\begin{aligned}\text{Consider, } \frac{s}{s^4 + s^2 + 1} &= \frac{s}{(s^2 + s + 1)(s^2 - s + 1)} = \frac{1}{2}\left[\frac{2s}{(s^2 + s + 1)(s^2 - s + 1)}\right] \\ &= \frac{1}{2}\left[\frac{(s^2 + s + 1) - (s^2 - s + 1)}{(s^2 + s + 1)(s^2 - s + 1)}\right] = \frac{1}{2}\left[\frac{1}{(s^2 - s + 1)} - \frac{1}{(s^2 + s + 1)}\right] \\ &= \frac{1}{2}\left[\frac{1}{\left(s - \frac{1}{2}\right)^2 + \frac{3}{4}} - \frac{1}{\left(s + \frac{1}{2}\right)^2 + \frac{3}{4}}\right]\end{aligned}$$

Therefore

$$\begin{aligned}L^{-1}\left\{\frac{s}{s^4 + s^2 + 1}\right\} &= \frac{1}{2}\left[e^{\frac{t}{2}}L^{-1}\left\{\frac{1}{s^2 + \frac{3}{4}}\right\} - e^{-\frac{t}{2}}L^{-1}\left\{\frac{1}{s^2 + \frac{3}{4}}\right\}\right] \\ &= \frac{1}{2}\left[e^{\frac{t}{2}}\frac{\sin \frac{\sqrt{3}}{2}t}{\frac{\sqrt{3}}{2}} - e^{-\frac{t}{2}}\frac{\sin \frac{\sqrt{3}}{2}t}{\frac{\sqrt{3}}{2}}\right] \\ &= \frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\sinh\left(\frac{t}{2}\right)\end{aligned}$$

Evaluation of $L^{-1}[e^{-as}F(s)]$

If $L[f(t)] = F(s)$, then $L[f(t - a)H(t - a)] = e^{-as}F(s)$

$$L^{-1}[e^{-as}F(s)] = f(t - a)H(t - a)$$

Examples

1. Evaluate: $L^{-1}\left\{\frac{e^{-5s}}{(s-2)^4}\right\}$.

Solution:

$$a = 5, F(s) = \frac{1}{(s-2)^4}$$

$$f(t) = L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{(s-2)^4}\right\} = e^{2t}L^{-1}\left\{\frac{1}{s^4}\right\} = \frac{e^{2t}t^3}{6}$$

$$L^{-1}\left\{\frac{e^{-5s}}{(s-2)^4}\right\} = f(t-a)H(t-a)$$

$$= \frac{e^{2(t-5)}(t-5)^3}{6} H(t-5).$$

2. Evaluate: $L^{-1}\left[\frac{e^{-\pi s}}{s^2+1} + \frac{se^{-2\pi s}}{s^2+4}\right]$.

Solution:

Given = $\{f_1(t-\pi)H(t-\pi) + f_2(t-2\pi)H(t-2\pi)\}$ (1)

$$f_1(t) = L^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$$

$$f_2(t) = L^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos 2t$$

(1) leads to

Given = $\sin(t-\pi)H(t-\pi) + \cos 2(t-2\pi)H(t-2\pi)$

$$= -\cos t H(t-\pi) + \cos(2t)H(t-2\pi).$$

Inverse transform of logarithmic and trigonometric functions

If $L[tf(t)] = F(s)$ then $L[tf(t)] = L^{-1}\left[-\frac{d}{ds}F(s)\right] = tf(t)$

In general, $L^{-1}\left[(-1)^n \frac{d^n}{ds^n}F(s)\right] = t^n f(t)$

Examples

1. Evaluate $L^{-1}\left\{\log\left(\frac{s+a}{s+b}\right)\right\}$.

Let $F(s) = \log\left(\frac{s+a}{s+b}\right) = \log(s+a) - \log(s+b)$

Then $-\frac{d}{ds}F(s) = -\left[\frac{1}{s+a} - \frac{1}{s+b}\right]$

$$L^{-1}\left[-\frac{d}{ds}F(s)\right] = -[e^{-at} - e^{-bt}]$$

Or $t f(t) = e^{-bt} - e^{-at}$. Thus $f(t) = \frac{e^{-bt} - e^{-at}}{b}$.

2. Evaluate $L^{-1}\left\{\tan^{-1}\frac{a}{s}\right\}$.

Let $F(s) = \tan^{-1}\left(\frac{a}{s}\right)$

Then $-\frac{d}{ds}F(s) = \left[\frac{a}{s^2 + a^2}\right]$ or $L^{-1}\left[-\frac{d}{ds}F(s)\right] = \sin at$ so that

or $t f(t) = \sin at, \quad f(t) = \frac{\sin at}{a}.$

Inverse transform of $\frac{F(s)}{s}$

Since $L\int_0^t f(t)dt = \frac{F(s)}{s}$, we have $L^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t f(t)dt$

Examples

1. Evaluate $L^{-1}\left[\frac{1}{s(s^2 + a^2)}\right].$

Solution:

Let $F(s) = \frac{1}{s^2 + a^2}$ so that $f(t) = L^{-1}F(s) = \frac{\sin at}{a}$

Then $L^{-1}\left[\frac{1}{s(s^2 + a^2)}\right] = L^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t \frac{\sin at}{a} dt = \frac{(1 - \cos at)}{a^2}$

2. $L^{-1}\left[\frac{1}{s^2(s+a)^2}\right].$

$L^{-1}\left\{\frac{1}{(s+a)^2}\right\} = te^{-at}$

$L^{-1}\left\{\frac{1}{s(s+a)^2}\right\} = \int_0^t e^{-at} t dt$
 $= \frac{1}{a^2} [1 - e^{-at}(1 + at)]$

Now $L^{-1}\left\{\frac{1}{s^2(s+a)^2}\right\} = \frac{1}{a^2} \int_0^t [1 - e^{-at}(1 + at)] dt$
 $= \frac{1}{a^3} [at(1 + e^{-at}) + 2(e^{-at} - 1)].$

Inverse Laplace transform of $F(s)$ using Convolution theorem

If $L^{-1}[F(s)] = f(t)$ and $L^{-1}[G(s)] = g(t)$, then

$$L^{-1}[F(s)G(s)] = \int_0^t f(u)g(t-u)du = f(t) * g(t)$$

This expression is called the convolution theorem for inverse Laplace transform.

Examples:

Employ convolution theorem to evaluate the following:

1. $L^{-1} \left\{ \frac{1}{(s+a)(s+b)} \right\}.$

Solution:

Let $F(s) = \frac{1}{s+a}, G(s) = \frac{1}{s+b}$

Taking inverse transform, we get $f(t) = e^{-at}, g(t) = e^{-bt}$

By convolution theorem,

$$\begin{aligned} L^{-1} \left\{ \frac{1}{(s+a)(s+b)} \right\} &= \int_0^t e^{-a(t-u)} e^{-bu} du = e^{-at} \int_0^t e^{(a-b)u} du \\ &= e^{-at} \left[\frac{e^{(a-b)t} - 1}{a-b} \right] = \frac{e^{-bt} - e^{-at}}{a-b}. \end{aligned}$$

2. $L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\}.$

Solution:

Let $F(s) = \frac{1}{s^2 + a^2}, G(s) = \frac{s}{s^2 + a^2}$, then $f(t) = \frac{\sin at}{a}, g(t) = \cos at$

By convolution theorem,

$$\begin{aligned} L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\} &= \int_0^t \frac{1}{a} \sin a(t-u) \cos au du \\ &= \frac{1}{a} \int_0^t \frac{\sin at + \sin(at-2au)}{2} du, \\ &= \frac{1}{2a} \left[u \sin at - \frac{\cos(at-2au)}{-2a} \right]_0^t = \frac{t \sin at}{2a}. \end{aligned}$$

(3) $L^{-1} \left\{ \frac{s}{(s-1)(s^2+1)} \right\}.$

Solution:

Let $F(s) = \frac{1}{s-1}, G(s) = \frac{s}{s^2+1}$

Then $f(t) = e^t, g(t) = \sin t$

$$\begin{aligned} L^{-1}\left\{\frac{1}{(s-1)(s^2+1)}\right\} &= \int_0^t e^{t-u} \sin u \, du = e^t \left[\frac{e^{-u}}{2} (-\sin u - \cos u) \right] \\ &= \frac{e^t}{2} [e^{-t} (-\sin t - \cos t) - (-1)] = \frac{1}{2} [e^t - \sin t - \cos t] \end{aligned}$$

Exercise:

By employing convolution theorem evaluate the following:

- | | |
|---|--|
| 1. $L^{-1} \left\{ \frac{1}{(s+1)(s^2+1)} \right\}$ | 4. $L^{-1} \left\{ \frac{s}{(s^2+a^2)(s^2+b^2)} \right\}, a \neq b.$ |
| 2. $L^{-1} \left\{ \frac{s}{(s+2)(s^2+9)} \right\}$ | 5. $L^{-1} \left\{ \frac{1}{s^2(s+2)^2} \right\}$ |
| 3. $L^{-1} \left\{ \frac{1}{(s^2+a^2)^2} \right\}$ | 6. $L^{-1} \left\{ \frac{4s+5}{(s-1)^2(s+2)} \right\}$ |

Answers:

$$\begin{aligned} \text{(ii)} \quad & \frac{1}{13} (2 \cos 3t + 3 \sin 3t - 2e^{-2t}) \quad \text{(iii)} \quad \frac{1}{2a^3} (\sin at - at \cos at) \quad \text{(iv)} \\ & \frac{1}{a^2 - b^2} (\cos bt - \cos at) \quad \text{(v)} \quad \frac{1}{4} (e^{-2t}(t+1) + t - 1) \end{aligned}$$

Laplace Transform Method for Differential Equations

As noted earlier Laplace transform technique is employed to solve initial-value problems. The solution of such a problem is obtained by using the Laplace Transform of the derivatives of function and then the inverse Laplace Transform.

The following are the expressions for the derivatives derived earlier.

$$L[f'(t)] = s L f(t) - f(0)$$

$$L[f''(t)] = s^2 L f(t) - s f(0) - f'(0)$$

$$L[f'''(t)] = s^3 L f(t) - s^2 f(0) - s f'(0) - f''(0)$$

Examples

1. $y'' + 2y' - 3y = \sin t, \quad y(0) = y'(0) = 0$

Solution:

Taking the Laplace transform of the given equation,

$$[s^2 Ly(t) - sy(0) - y'(0)] + 2[s Ly(t) - y(0)] - 3 L y(t) = \frac{1}{s^2+1}$$

Using the given conditions, we get

$$[s L\{y(t)\} - y(0)] + L\{y(t)\} = \frac{1}{(s+1)^2}$$

Using the given condition, this becomes

$$(s+1)L\{y(t)\} - 2 = \frac{1}{(s+1)^2} \text{ so that } L\{y(t)\} = \frac{2s^2 + 4s + 3}{(s+1)^3}$$

Taking the inverse Laplace transform, we get

$$\begin{aligned} Y(s) &= L^{-1} \left\{ \frac{2s^2 + 4s + 3}{(s+1)^3} \right\} \\ &= L^{-1} \left[\frac{2(s+1-1)^2 + 4(s+1-1) + 3}{(s+1)^3} \right] \\ &= L^{-1} \left[\frac{2}{s+1} + \frac{1}{(s+1)^3} \right] = \frac{1}{2} e^{-t} (t^2 + 4) \\ L[y(t)] [s^2 + 2s - 3] &= \frac{1}{s^2 + 1} \text{ or } L[y(t)] = \frac{1}{(s-1)(s+3)(s^2+1)} \text{ or} \\ y(t) &= L^{-1} \left[\frac{1}{(s-1)(s+3)(s^2+1)} \right] = L^{-1} \left[\frac{A}{s-1} + \frac{B}{s+3} + \frac{Cs+D}{s^2+1} \right] \\ &= L^{-1} \left[\frac{1}{8} \frac{1}{s-1} - \frac{1}{40} \frac{1}{s+3} + \frac{-\frac{s}{10} - \frac{1}{5}}{s^2+1} \right] \text{ (method of partial fractions)} \\ &= \frac{1}{8} e^t - \frac{1}{40} e^{-3t} - \frac{1}{10} (\cos t + 2 \sin t). \end{aligned}$$

2. Employ Laplace Transform method to solve the integral equation.

$$f(t) = 1 + \int_0^t f(u) \sin(t-u) du$$

Solution:

Taking Laplace transform of the given equation, we get

$$L f(t) = \frac{1}{s} + L \int_0^t f(u) \sin(t-u) du$$

By using convolution theorem here

$$L f(t) = \frac{1}{s} + L f(t) \cdot L \sin t = \frac{1}{s} + \frac{L f(t)}{s^2 + 1} \text{ Thus}$$

$$L f(t) = \frac{s^2 + 1}{s^3} \quad \text{or} \quad f(t) = L^{-1} \left(\frac{s^2 + 1}{s^3} \right) = 1 + \frac{t^2}{2}.$$

3. A particle is moving along a path satisfying the equation $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 0$ where x denotes the displacement of the particle at time t . If the initial position of the particle is at $x = 20$ and the initial speed is 10, find the displacement of the particle at any time t using Laplace transforms.

Solution:

Given equation may be rewritten as

$$x''(t) + 6x'(t) + 25x(t) = 0$$

Here the initial conditions are $x(0) = 20$, $x'(0) = 10$.

Taking the Laplace transform of the equation, we get

$$L\{x(t)\}[s^2 + 6s + 25] - 20s - 130 = 0 \quad \text{or} \quad L\{x(t)\} = \frac{20s + 130}{s^2 + 6s + 25}$$

so that

$$\begin{aligned} x(t) &= L^{-1} \left[\frac{20s + 130}{(s + 3)^2 + 16} \right] = L^{-1} \left[\frac{20(s + 3) + 70}{(s + 3)^2 + 16} \right] \\ &= 20 L^{-1} \frac{s + 3}{(s + 3)^2 + 16} + 70 L^{-1} \frac{1}{(s + 3)^2 + 16} \\ &= 20 e^{-3t} \cos 4t + 35 \frac{e^{-3t} \sin 4t}{2}. \end{aligned}$$

4. A voltage Ee^{-at} is applied at $t = 0$ to a circuit of inductance L and resistance R . Show that the

$$\text{current at any time } t \text{ is } \frac{E}{R - aL} \left[e^{-at} - e^{-\frac{Rt}{L}} \right].$$

Solution:

The circuit is an LR circuit. The differential equation with respect to the circuit is

$$L \frac{di}{dt} + Ri = E(t)$$

Here, L denotes the inductance, i denotes current at any time t and $E(t)$ denotes the E.M.F.

It is given that $E(t) = E e^{-at}$. With this, we have

Thus, we have

$$L \frac{di}{dt} + Ri = Ee^{-at} \text{ or } Li'(t) + R i(t) = Ee^{-at}$$

Taking Laplace transform on both sides to get

$$L[L\{i'(t)\}] + R[L\{i(t)\}] = E L\{e^{-at}\}$$

$$L[s L\{i(t)\} - i(0)] + R[L\{i(t)\}] = E \frac{1}{s+a}$$

$$\text{Since } i(0) = 0, \quad L\{i(t)\}[sL + R] = \frac{E}{s+a} \quad \text{or}$$

$$L\{i(t)\} = \frac{E}{(s+a)(sL+R)}$$

$$\text{Taking inverse transform } L^{-1}\{i(t)\} = L^{-1}\left\{\frac{E}{(s+a)(sL+R)}\right\}$$

$$= \frac{E}{R-aL} \left[L^{-1}\left\{\frac{1}{s+a}\right\} - L^{-1}\left\{\frac{1}{sL+R}\right\} \right]$$

$$\text{Thus } i(t) = \frac{E}{R-aL} \left[e^{-at} - e^{-\frac{Rt}{L}} \right].$$

5. Mass spring damper system can be modeled using Newton's and Hooke's law. The differential equation representing this system is $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 4\sin\omega t$ with initial conditions $x(0) = x'(0) = 0$.

Solution:

Given equation may be rewritten as

$$x''(t) + 6x'(t) + 25x(t) = 4\sin\omega t$$

Here the initial conditions are $x(0) = 0, x'(0) = 0$.

Taking the Laplace transform of the equation, we get

$$L\{x(t)\}[s^2 + 6s + 25] = \frac{4\omega}{s^2 + \omega^2} \quad \text{or } L\{x(t)\} = \frac{4\omega}{(s^2 + \omega^2)(s^2 + 6s + 25)}$$

On resolving in to partial fractions (with $\omega = 2$) leads to

$$\begin{aligned} x(t) &= L^{-1} \left[\frac{8}{(s^2 + 4)(s^2 + 6s + 25)} \right] = L^{-1} \left[\frac{As + B}{s^2 + 4} + \frac{Cs + D}{s^2 + 6s + 25} \right] \\ &= \frac{4}{195} [7 \sin 2t - 4 \cos 2t] + \frac{2}{195} [e^{-3t} (8 \cos 4t - \sin 4t)]. \end{aligned}$$

Exercise:

Employ Laplace transform method to solve the following initial value problems.

$$(i) \quad y'' + 5y' + 6y = e^{-2t}, \quad y(0) = y'(0) = 1$$



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(ii) $y''' + 2y'' - y' - 2y = 0, \quad y(0) = 1, \quad y'(0) = 2 = y''(0)$

(iii) $y'' + 2y' + 5y = e^{-t} \sin t, \quad y(0) = 0, \quad y'(0) = 1.$

Answers:

(i) $3e^{-2t} - 2e^{-3t} + te^{-2t}$ (ii) $\frac{1}{3}e^{-2t} + \frac{5}{3}e^{-t} - e^t$ (iii) $\frac{1}{3}e^{-t}(\sin t + \sin 2t)$

Video links:

<https://www.youtube.com/watch?v=6MXMDrs6ZmA>

<https://www.youtube.com/watch?v=wnnnv4wt-Lw>

<https://www.youtube.com/watch?v=N-zd-T17uiE>

<https://www.youtube.com/watch?v=l7nzLD3t4Uc>